ITPM

IT Project Management

Unit 1 — Project Overview & Project Scheduling

Simple, Easy-to-Understand Notes

1. Project Overview

A project overview is a short, complete summary of a project — software or system — written before the work begins.

It answers 3 basic questions:

- What is the project?
- Why is it needed (the cause)?
- What benefit will it give?

5 Parts of a Project Overview

#	Part	What it means
1	Project Name	Title of the project
2	Project Description	Details of what the project is about
3	Objective	The reason the project is being made
4	Scope	Who will use the project (its users/boundary)
5	Expected Outcome	The result / feedback expected at the end

2. Feasibility Study

A feasibility study checks whether a project is actually possible to do and whether it will be profitable, before we spend time and money on it.

Example: For a college ERP system, before building it we check — What is the budget? Who are the developers? How much time will it take? Should the software even be created?

4 Types of Feasibility Study

Type	Checks	Key Questions
1. Technical Feasibility	Whether the technology needed is available	What hardware, software, internet, programming language and database are needed?
2. Economic Feasibility	Whether the project will be profitable	What is the project cost? Will it give profit?
3. Operational Feasibility	Whether the system will work smoothly for people	Will staff use it easily? Do they need training? Will users be satisfied?
4. Schedule Feasibility	Whether the project can be completed on time	Can it be finished in the decided time? (e.g., decided 3 months → completed in 3 months)

💡 *Easy way to remember: Technical = Technology, Economic = Profit, Operational = People/Staff use, Schedule = Time.*

3. Identification & Its Steps

Identification means finding out the real problem, and checking whether a software solution is actually needed to solve it.

8 Steps of Identification

Step	Name	Meaning
1	Identify the problem	e.g. "No online attendance system" — taking attendance physically takes time
2	Analyze current situation	Study how things are currently being done
3	Identify stakeholders	Who is affected/involved — e.g. teachers, students
4	Gather requirements	Take interviews, surveys, meetings to understand the problem
5	Define project objective	Set a clear goal/objective
6	Identify possible solutions	e.g. Mobile App, Web Application
7	Preliminary feasibility check	Quick check of Technical, Schedule, Operational, Economic feasibility
8	Prepare project proposal	Report stating whether the project is needed or not

4. Market & Demand Analysis

Before starting an IT project, we must study the market — customers, competitors, and how many competitors exist — and estimate the future demand for the product.

4.1 Market Analysis

Market Analysis = studying the overall market environment (the entire market).

- Market growth
- Industry trends
- Target customers
- Pricing strategy

Example: Food Delivery App → check number of online users, market trends, and existing food delivery services.

4.2 Demand Analysis

Demand Analysis finds out how many customers will buy or use the product.

- Customer needs
- Product price
- Population
- Seasonal demand
- Income level

Example: Online Learning Platform → check number of students, internet availability, and demand for online courses.

4.3 Steps to Perform Market & Demand Analysis

Step	Name	Details
1	Identify the target market	Who will use the product?
2	Collect market information	Surveys, interviews, online search
3	Analyze customer needs	Customer expectations, pain points, desired features
4	Analyze competitors	What product they offer, their pricing, their weaknesses
5	Estimate market demand	Number of potential customers, future growth, expected sales
6	Forecast future demand	Use prediction, historical data, customer feedback

Step	Name	Details
7	Prepare report	Market size, target market, risks, business opportunities

5. Project Cost Estimation

Project cost estimation is the process of predicting the total cost required to complete an IT project.

Objectives

1. Determine the total project budget
2. Supports project planning & scheduling
3. Allocate resources properly
4. Reduce financial risks

5.1 Components of Project Cost Estimate

Component	Includes
1. Human Resource Cost	Project management, software developer, tester, UI/UX designer, business analyst
2. Hardware (H/W) Cost	Computer, servers, network devices, printers
3. Software (S/W) Cost	Software licenses, OS, databases, antivirus
4. Training Cost	Training employees on new technology, workshops, instructor fees, online certificates
5. Infrastructure Cost	Internet, electricity, office rent, furniture (in addition to computer/server/network/printer)

6. Financial Appraisal

Financial appraisal is the process of checking whether the software project will actually give profit and is financially feasible — it evaluates the reason/worth of the project.

Objectives

5. To estimate the total project cost
6. To estimate profit
7. To compare different investment options
8. To minimize financial risks

Components

- Initial Investment — amount of money needed to start the project
- Hardware (H/W) Cost
- Software (S/W) Cost
- Office Setup
- Employee Training
- Operating Cost — repeated expenses during the project (salary, electricity bills, maintenance, internet charges)
- Revenue Estimation — expected income generated from the project (product sales, service charges)
- Cash Flow Analysis — movement of money IN (cash inflow/income) and OUT (cash outflow/expenses)

6.1 Financial Appraisal Techniques

1. Payback Period (PBP)

Tells us how many years it takes to get back (recover) the money we invested.

PBP = Initial Investment ÷ Annual Cash Inflow

2. Return on Investment (ROI)

ROI = (Net Profit ÷ Investment) × 100

3. Benefit Cost Ratio (BCR)

BCR = Present Value of Benefit ÷ Present Value of Cost

Advantages of Financial Appraisal

9. Helps in investment decisions
10. Measures profit
11. Reduces financial risk

7. Project Scheduling

Project scheduling is the process of creating a timetable (IT) for all project activities — deciding their sequence, duration, resources, and start/end dates.

Objectives

12. Complete the project on the planned deadline
13. Proper use of resources
14. Monitor project progress
15. Reduce project risk

7.1 Steps in Project Scheduling

Step	Name	Meaning
1	Define project activities	Divide the project into small activities/tasks
2	Sequence activities	Determine/arrange the order of activities
3	Estimate activity duration	Find out the time each activity takes to perform
4	Assign resources	Allocate team members, hardware, and software
5	Develop the project schedule	Using tools like Gantt Chart, PERT, CPM (Critical Path Method)
6	Monitor and update	Track the progress regularly

Benefits of Project Scheduling

16. Timely completion of projects
17. Better monitoring
18. Improves project planning

8. Critical Path Method (CPM)

The Critical Path is the longest path of activities in a project — it decides the minimum total time needed to complete the whole project. An activity that does not depend on any other activity is the one we start with.

Important Terms

Term	Full Form
ES	Early Start
EF	Early Finish
D	Duration
LS	Late Start
LF	Late Finish
SL	Slack Value (Float)

Node Box Format used in diagrams

Each activity node is written in a box like this:

ES	D	EF
LS	SL	LF

Key Formulas

EF = ES + Duration

SL = LF − EF

LS = LF − D

How to Solve (2 Passes)

- Forward Pass → calculate ES and EF (start from the beginning, take the MAXIMUM value where paths join)
- Backward Pass → calculate LS and LF (start from the end, take the MINIMUM value where paths join)

💡 *The Critical Path is chosen as the path where the Slack Value (SL) = 0 for every activity on it.*

Worked Example

Activities and durations (in months):

Activity	Predecessor	Duration
A	—	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

After doing the forward pass (max value) and backward pass (min value), the path with Slack = 0 is found.

Critical Path = A → B → D → F

9. Float Calculation

Float (also called Slack) tells us how much an activity can be delayed without delaying the whole project. Float is calculated using the Critical Path.

Float (F) = Latest − Earliest

Types of Float

Type	Formula	Meaning
Total Float	LS − ES	Total time an activity can be delayed without delaying the project end date
Free Float	Ej − Ei − tij	Delay allowed without affecting the next activity's early start
Independent Float	Ej − Li − tij	Delay allowed without affecting activities before or after it

💡 *Ej = Early start of the last (ending) node. Li = Late start of the first (starting) node. Ei = Early start of the first node.*

💡 *Rule: Where Total Float = 0, that path is the Critical Path.*

10. PERT vs CPM

PERT (Program Evaluation & Review Technique) and CPM (Critical Path Method) are both used for project scheduling, but they differ in the way they estimate time.

Basis	PERT	CPM
Nature	Probabilistic (finds an estimated value, may vary)	Deterministic (finds one exact value/duration)
Time Estimation	Uses 3 time estimates: Optimistic (to), Most Likely (tm), Pessimistic (tp)	Uses a single time duration
Focus	Time management	Cost control
Suitable For	Development / research-type projects	Manufacturing / repetitive projects
Orientation	Event-oriented	Activity-oriented
Cost Consideration	Time is important, cost is not the main focus	Cost is important
Expected Time Formula	Te = (to + 4tm + tp) / 6	No such formula — uses a fixed time
Critical Path	Can vary (due to change in situation)	Fixed

PERT Formulas

Expected Time: $Te = (to + 4tm + tp) / 6$

Variance: $\sigma^2 = ((tp - to) / 6)^2$

Where: to = Optimistic time, tm = Most Likely time, tp = Pessimistic time.

Steps to Solve a PERT Problem

19. Draw the project network diagram (based on activities)
20. Find Expected Duration (Te) & Variance (σ^2) for every activity
21. Find Earliest and Latest occurrence at every node (forward pass = largest value; backward pass = smallest value)
22. Find the Expected Project Length (Critical Path) — the path where the difference between Latest and Earliest occurrence is 0

11. Precedence Relationship

A precedence relationship shows how two activities are linked/dependent on each other. There are 4 types:

#	Type	Meaning
1	Finish to Start (FS)	First activity must finish before the next one starts
2	Start to Start (SS)	Both activities start together
3	Finish to Finish (FF)	Both activities must finish together
4	Start to Finish (SF)	First activity must start before the second one can finish

12. Crashing

Crashing is a project management technique used to complete a project faster by adding extra resources to critical activities. It is mainly used for cost reduction / time reduction in the CPM method.

Key Points

- Reduces project completion time
- Focuses only on critical path activities
- Increases direct cost (extra staff, overtime)
- Decreases indirect cost (administration, office expenses)

Steps of Crashing

23. Identify the Critical Path
24. Find activities that can be crashed
25. Calculate the cost of crashing
26. Crash the least expensive critical activity first
27. Repeat until the target time is achieved

Advantages

- Finishes the project earlier
- Reduces overall cost (indirect)
- Helps meet deadlines

Disadvantages

- Increases direct cost
- Not all activities can be crashed

Quick Revision Summary

Topic	One-line Takeaway
Feasibility Study	4 types: Technical, Economic, Operational, Schedule
Identification	8 steps — from identifying the problem to preparing a project proposal
Market Analysis	Studies the overall market (growth, trends, customers, pricing)
Demand Analysis	Studies how many customers will buy/use the product
Cost Estimation	5 components: Human Resource, H/W, S/W, Training, Infrastructure
Financial Appraisal	3 techniques: Payback Period, ROI, Benefit Cost Ratio
Project Scheduling	6 steps — from defining activities to monitoring progress

Topic	One-line Takeaway
CPM	Finds the Critical Path using Forward Pass (max) and Backward Pass (min); Slack = 0
Float	Total, Free, Independent — shows how much delay is allowed
PERT	Uses 3 time estimates (Optimistic, Most Likely, Pessimistic) — probabilistic
Precedence Relationship	4 types: FS, SS, FF, SF
Crashing	Adds extra resources to critical activities to finish the project faster